

2. Uniform boundedness principle:

$T_n : X \rightarrow Y$ be bdd linear functions with X be Banach space & Y be any NLS.

Now, if $\sup_{n \in \mathbb{N}} \|T_n(x)\|_Y < \infty$, $M_n > 0$. ✓

✓ $\Rightarrow \sup_{n \in \mathbb{N}} \|T_n\| < M$, for some $M > 0$.

We are given that $\forall \xi \in \mathcal{H}$ & $\forall \eta \in \mathcal{K}$

$\{\langle T_n \xi, \eta \rangle\}_{n \in \mathbb{N}}$ is Cauchy seq.

$$\Rightarrow |\langle T_n \xi, \eta \rangle| \leq \|T_n \xi\|_K \|\eta\|_K \quad \text{--- } (*)$$

by C-S ineqⁿ.

$$\checkmark \{ \langle T_n \xi, T_n \xi \rangle \}_{n \in \mathbb{N}} \text{ is Cauchy seqⁿ}$$

$$\Rightarrow \{ \|T_n \xi\|^2 \} \quad \text{--- } \dots$$

$$\Rightarrow \{ \|T_n \xi\| \} \quad \text{--- } \dots \quad \text{--- } \# \checkmark$$

$\checkmark$ [Example: $\{-1, 1\}$ is not Cauchy, but $\{-1, 1\}^2 = \{1, 1\}$ is Cauchy]

$T_n: H \rightarrow K$ bdd linear fm:

We can U.B.P. to the seqⁿ $\{T_n\}$.

$$\boxed{\sup_{n \in \mathbb{N}} \|T_n \xi\| < M_\xi, \exists M_\xi > 0} \quad \checkmark$$

$$\Rightarrow \boxed{\sup_{n \in \mathbb{N}} \|T_n\| < \underline{M} < \infty. \exists \underline{M} > 0} \quad \checkmark$$

Prove $(\#)$,

Given $\{a_n^2\}$ is Cauchy where $a_n \geq 0 \forall n \in \mathbb{N}$
 s.t. $\{a_n\}$ is Cauchy.

$$|a_m^2 - a_n^2| < \varepsilon \quad \forall n, m \geq N,$$

$$\Rightarrow |(a_m - a_n)(a_m + a_n)| < \varepsilon$$

$$\Rightarrow |a_m - a_n| |a_m + a_n| < \varepsilon$$

$$\Rightarrow |a_m - a_n| < \frac{\varepsilon}{|a_m + a_n|} < \frac{\varepsilon}{|a_n|}$$

$$|a_n^2| < M$$

$$[\because a_m \geq 0]$$

$$\Rightarrow |a_n|^2 < M \Rightarrow |a_n| < \sqrt{M}$$

$$\text{Now, } \underline{a_{n-1}}, a_n \rightarrow 0.$$

$$\Rightarrow |a_n|^2 > m$$

$$\Rightarrow |a_n| > \sqrt{m} \Rightarrow \{a_n\} \text{ is Cauchy.}$$

$$\text{Cauchy-III: } a_n^2 \rightarrow 0$$

$$\forall \varepsilon, \exists N_\varepsilon \in \mathbb{N} \text{ s.t.}$$

$$\forall n \geq N_\varepsilon \Rightarrow |a_n^2| < \varepsilon^2$$

$$\Leftrightarrow |a_n|^2 < \varepsilon^2$$

$$\Leftrightarrow |a_n| < \varepsilon \Rightarrow a_n \rightarrow 0 \Rightarrow \{a_n\} \text{ is Cauchy.}$$

(b) Riesz-Representation Theorem:

$T: X \rightarrow K$ [$K = \mathbb{C}$ or $\mathbb{R}$], be bounded linear operator & X be a Hilbert space, then

$$\exists t \in X \text{ s.t. } T_n := \langle x, t \rangle$$

b.f.

$$F_\zeta(\eta) = \lim_{n \rightarrow \infty} \langle \eta, T_n \zeta \rangle$$

$$\begin{aligned} F_\zeta(a\eta_1 + b\eta_2) &= \lim_{n \rightarrow \infty} \langle a\eta_1 + b\eta_2, T_n \zeta \rangle \\ &= aF_\zeta(\eta_1) + bF_\zeta(\eta_2) \end{aligned}$$

$$\begin{aligned} |F_\zeta(\eta)| &= \lim_{n \rightarrow \infty} |\langle \eta, T_n \zeta \rangle| \\ &\leq \lim_{n \rightarrow \infty} (\|\eta\| \|T_n \zeta\|) \text{ [by C-S inequality]} \\ &= \left(\lim_{n \rightarrow \infty} \|T_n \zeta\| \right) \|\eta\| \\ &= M_\zeta \|\eta\| \longrightarrow (\text{From } \textcircled{*}) \end{aligned}$$

$$|F_\zeta(\eta)| \leq M_\zeta \|\eta\| \Rightarrow F_\zeta \text{ is bounded linear functional.}$$

By R-P-Theorem, $\textcircled{\exists!} \eta_\zeta \in K$ s.t.

$$F_\zeta(\eta) = \langle \eta, \eta_\zeta \rangle$$

Define $T: H \rightarrow K$ it is well defined by the
 $\xi \mapsto \eta_\xi$ uniqueness of η_ξ for
each ξ

$$\overline{\langle \eta, T\xi \rangle} = \overline{\langle \eta, \eta_\xi \rangle} = \overline{F_\xi(\eta)}$$

$$= \lim_{n \rightarrow \infty} \langle \eta, T_n \xi \rangle$$

$$\boxed{\langle T\xi, \eta \rangle = \lim_{n \rightarrow \infty} \langle T_n \xi, \eta \rangle} = \lim_{n \rightarrow \infty} \langle \eta, T_n \xi \rangle$$

$$\langle T(a\xi_1 + b\xi_2), \eta \rangle$$

$$= \lim_{n \rightarrow \infty} \langle T_n(a\xi_1 + b\xi_2), \eta \rangle$$

$$= \langle aT(\xi_1) + bT(\xi_2), \eta \rangle \quad \forall \eta \in K$$

$$\Rightarrow T(a\xi_1 + b\xi_2) = aT(\xi_1) + bT(\xi_2)$$

$$\|T(\xi)\|^2 = \langle T\xi, T\xi \rangle$$

$$= \lim_{n \rightarrow \infty} \langle T_n \xi, T\xi \rangle$$

$$\leq \lim_{n \rightarrow \infty} \|T_n \xi\| \|T\xi\|$$

$$\leq M \| \zeta \| \quad \underline{\| T \zeta \|}$$

$$\| T_n \zeta \| \leq \| T_n \| \| \zeta \|$$

$$\leq \left(\sup_{n \in \mathbb{N}} \| T_n \| \right) \| \zeta \|$$

$$\text{dt } \| T_n \zeta \| \leq M \| \zeta \|$$

$n \rightarrow \infty$

$$\Rightarrow \boxed{\| T \zeta \| \leq M \| \zeta \|} \Rightarrow T \text{ is linear and } \underline{\text{bounded.}}$$

Result: $f: X \longrightarrow \{ T: X \rightarrow \mathbb{K} \mid T \text{ is bounded \& linear} \}$

$\underbrace{\hspace{10em}}_{X^*}$

$\boxed{x \longmapsto \langle -, x \rangle}$

Show that f is bijjective.

pf: $\langle -, x \rangle = \langle -, y \rangle$

$$\Rightarrow \langle \eta, x \rangle = \langle \eta, y \rangle \quad \forall \eta \in X$$

$$\Rightarrow \langle \eta, x - y \rangle = 0 \quad \forall \eta \in X$$

$$\Rightarrow \textcircled{x=y} \cdot \boxed{\text{taking } \eta = x - y} \Rightarrow f \text{ is 1-1.}$$

By R.P. Th^m. f is [onto]
 $\Rightarrow f$ is bijective.

$$X \cong X^*$$

$\exists f \text{ } K = \mathbb{R} \Rightarrow f \text{ is the linear isomorphism}$
 $X \cong X^*$

But if $K = \mathbb{C}$ (?)